

Diffusion Tensor Imaging “DTI” in Primary Open Angle Glaucoma

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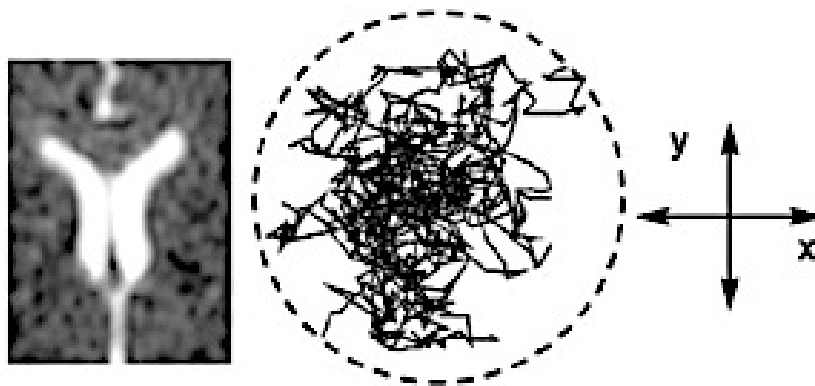
No disclosures

- ❖ Glaucoma is considered the third leading cause of blindness with over 8 million new cases each year.
- ❖ It is a disease that is characterized by retinal ganglionic cell death associated with reduction in the total number of axonal fibers that make up the optic nerve.
- ❖ High intraocular pressure (IOP) is one of the most consistent risk factors associated with primary open angle glaucoma (POAG) however, the concept that high IOP is a defining characteristic of POAG has been almost discarded as retinal ganglionic cell degeneration continues after the intraocular pressure is normalized.

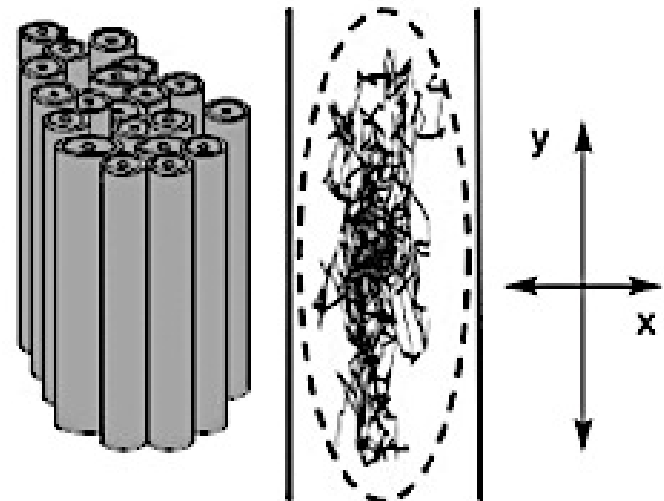
DTI

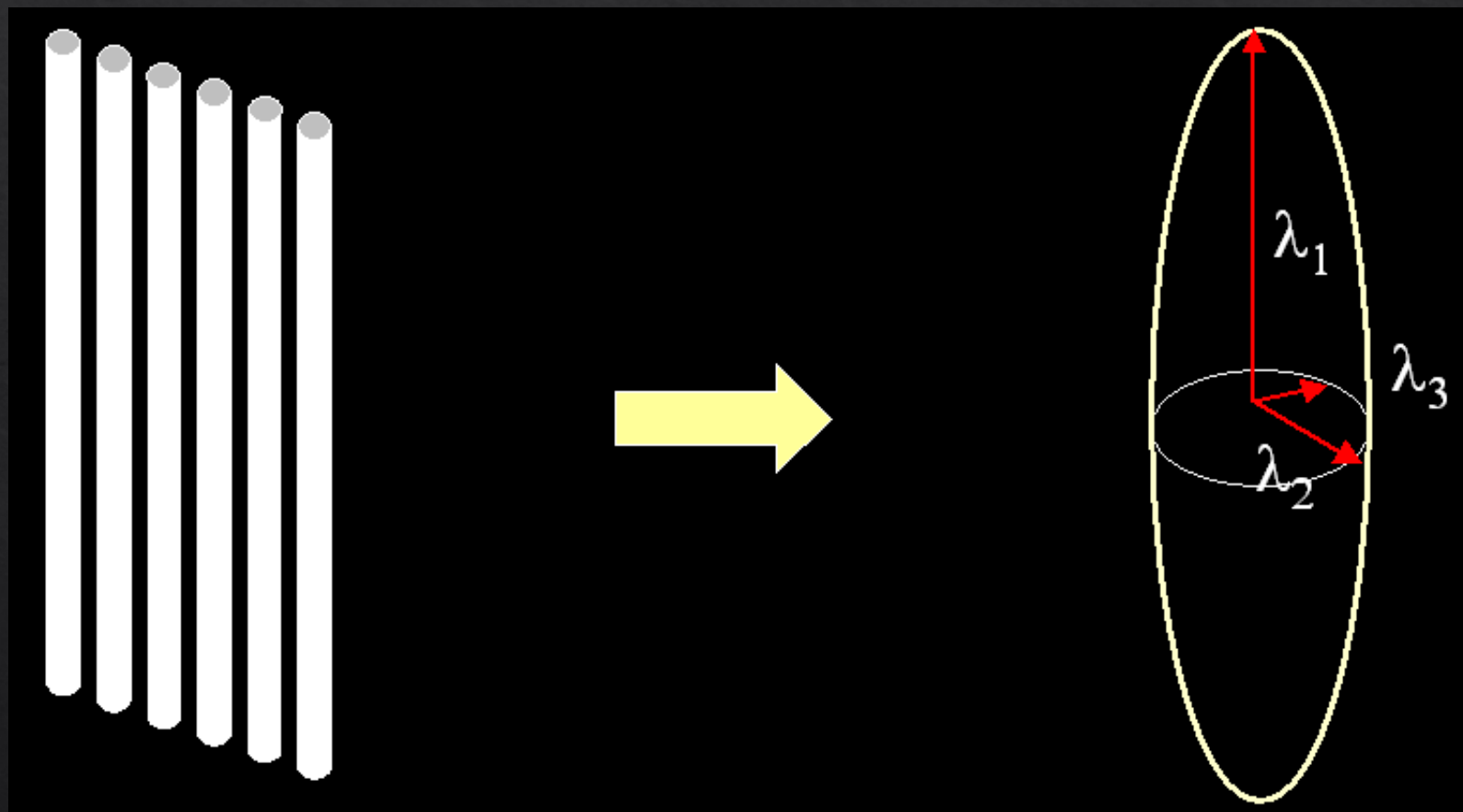
Diffusion Tensor Imaging

A. Isotropic Diffusion



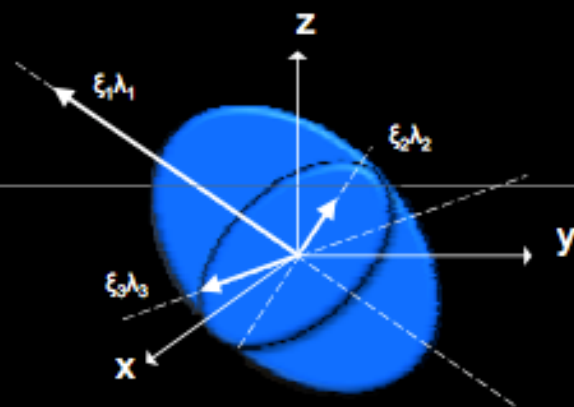
B. Anisotropic Diffusion





Diffusion Tensor Imaging

$$D = \begin{bmatrix} D_{xx} & D_{xy} & D_{xz} \\ D_{yx} & D_{yy} & D_{yz} \\ D_{zx} & D_{zy} & D_{zz} \end{bmatrix}$$



$$FA = \frac{\sqrt{3}}{\sqrt{2}} \frac{\sqrt{(\lambda_1 - \lambda)^2 + (\lambda_2 - \lambda)^2 + (\lambda_3 - \lambda)^2}}{\sqrt{\lambda_1^2 + \lambda_2^2 + \lambda_3^2}}$$

Fractional Anisotropy = 0

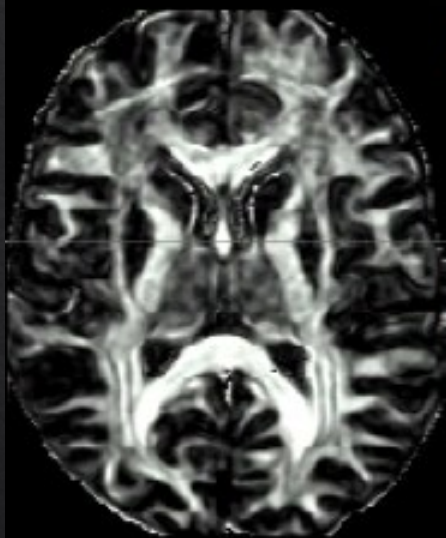
= 1



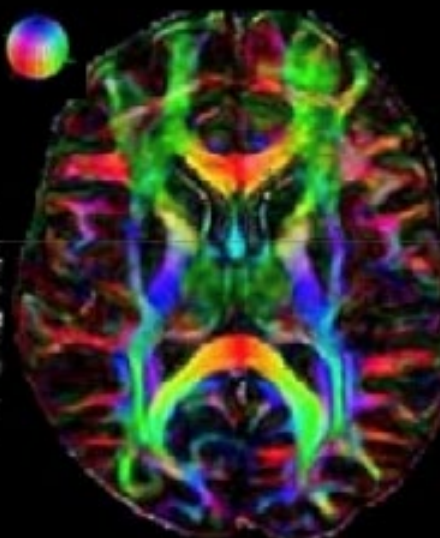
$$RA = \frac{\sqrt{3}}{\sqrt{2}} \frac{\sqrt{(\lambda_1 - \lambda)^2 + (\lambda_2 - \lambda)^2 + (\lambda_3 - \lambda)^2}}{\lambda_1 + \lambda_2 + \lambda_3}$$

Mean Diffusivity (ADC) = $(\lambda_1 + \lambda_2 + \lambda_3)/3$

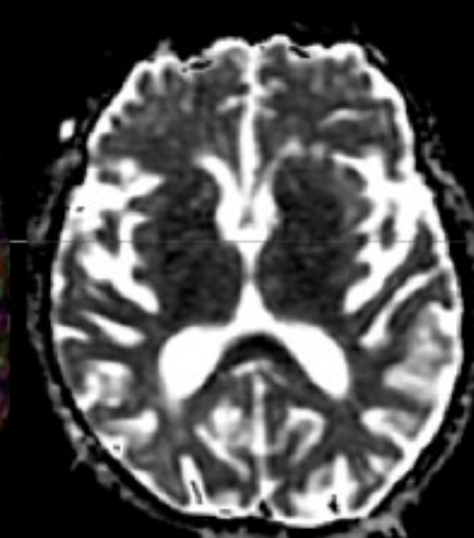
Directionally Encoded Fractional Anisotropy



FA

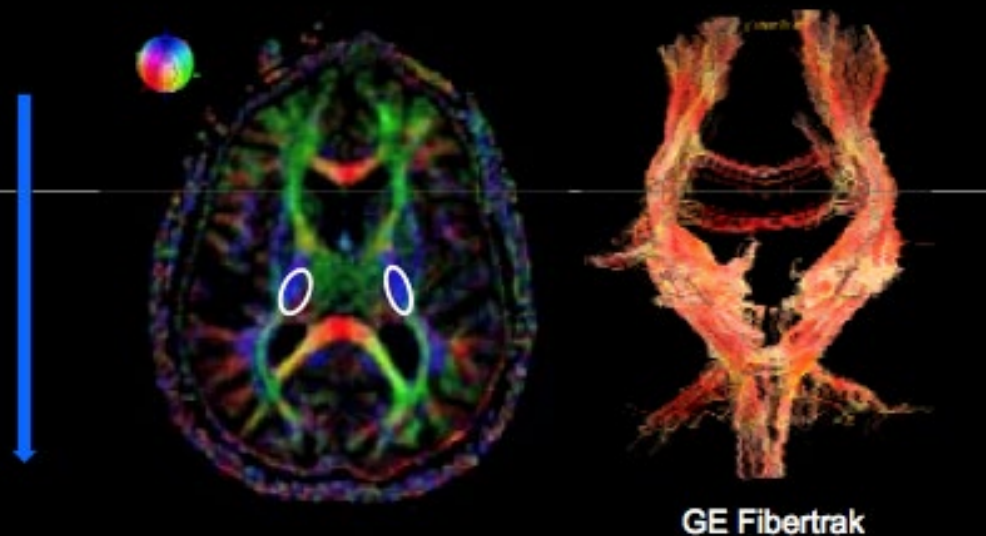


Color FA



ADC

Standard Tractography



- ◆ To evaluate Diffusion Tensor Imaging (DTI) findings at 7 different segments of the visual pathway (from the intra-orbital optic nerve to the visual cortex) to detect spread of glaucomatous damage into the nervous system in POAG patients.
- ◆ To correlate fractional anisotropy (FA) values of the visual pathway with clinical and quantitative ophthalmologic parameters including the severity and duration of disease, intraocular pressure and thickness of the retinal nerve fiber layer (RNFL).

- ◆ Case/Control study on thirty individuals: (15 patients and 15 age/gender-matched controls) were included.
- ◆ Patients (9 males and 6 females with an age range from 31 to 59 years, median age= 50) were referred by our ophthalmology outpatient clinic during the time period from December 2015 to April 2016 with a clinical diagnosis of POAG.

◇ All subjects (n = 30) underwent MRI of the orbit and brain (1.5 Tesla scanner) including:

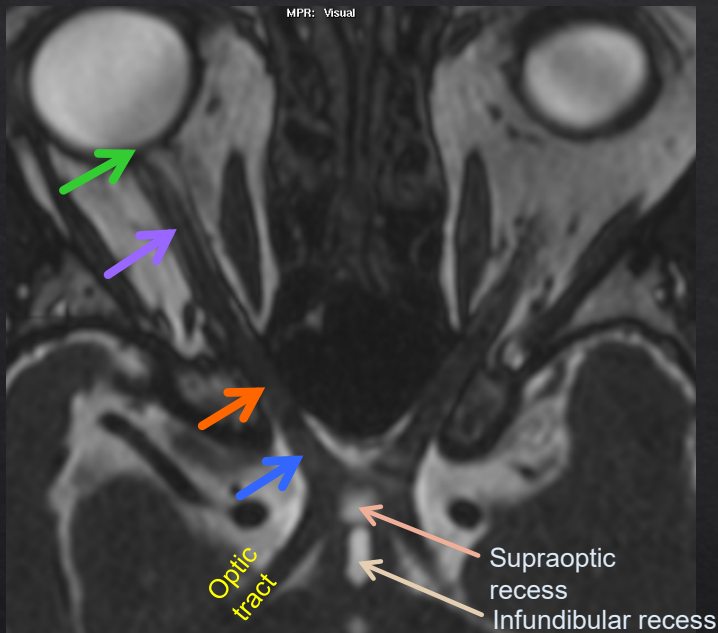
1) T1MPRAGE, T2 and FLAIR sequences.

2) Semi-automated segmentation and quantification of fiber tracts at seven different regions-of-interest (ROIs) that were used for the evaluation of fractional anisotropy in patients with glaucoma and controls along the right and left visual pathways (figure 1a, 1b, 1c) identified on consensus reading by two expert neuroradiologists as follows:

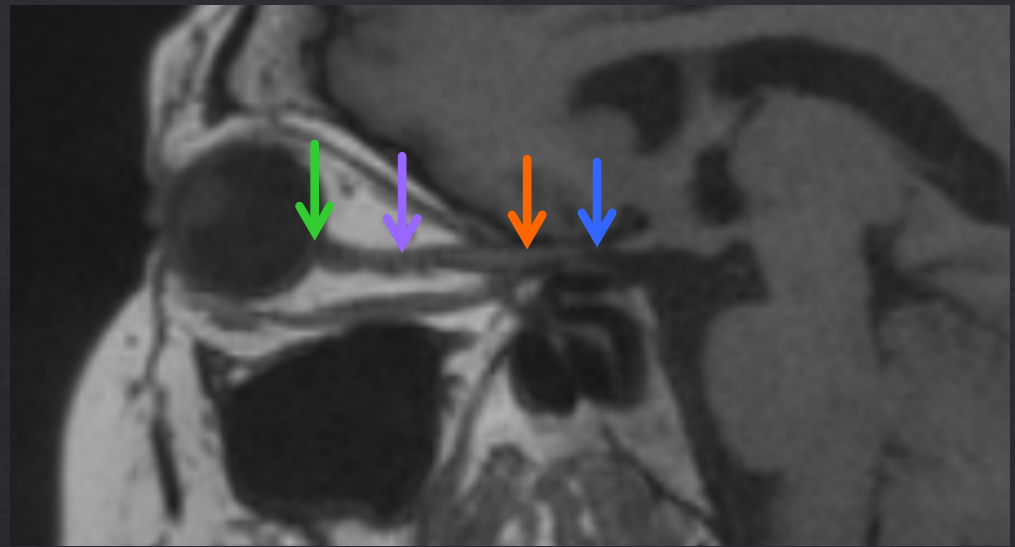
ROI 1: Intraorbital part of the optic nerve, ROI 2: Intracranial part of the optic nerve, ROI 3: Optic chiasm, ROI 4: Lateral geniculate nucleus LGN, and three separate points along the optic radiation: ROI 5: directly after/posterior to LGN, ROI 6: at the level of occipital horn of lateral ventricle, and ROI 7: right before optic fibers cortical spread.

CN 2

Optic Nerve



Axial CISS of CN 2



Sagittal oblique T1 SPACE CN 2

- Four segments: **retinal**, **orbital**, **cannalicular**, and **cisternal**.
- **Retinal segment** leaves globe via the optic foramen of the sclera.
- **Orbital segment** is surrounded by dural sheath and CSF.
- **Cannalicular** segment lies in the optic canal, below ophthalmic artery.
- **Cisternal** segment lies in the suprasellar cistern, inferomedial to the anterior cerebral artery.
- CN2 decussates and terminates at the optic chiasm, giving rise to **optic tracts**.

CLICK TO PAUSE/RESTART VIDEO

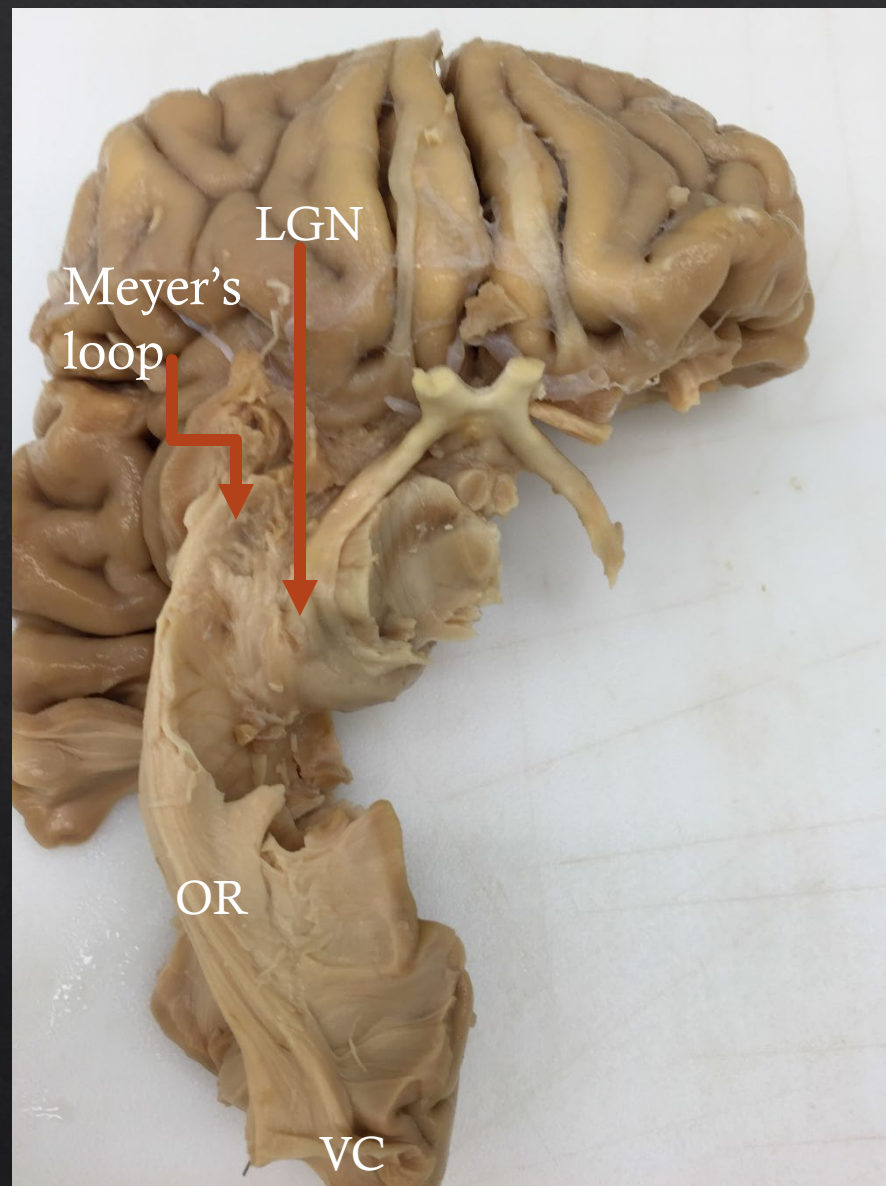


Figure 1a: ROI 1: intraorbital part of the optic nerve, ROI 2: intracranial part of the optic nerve, ROI 3: optic chiasm.

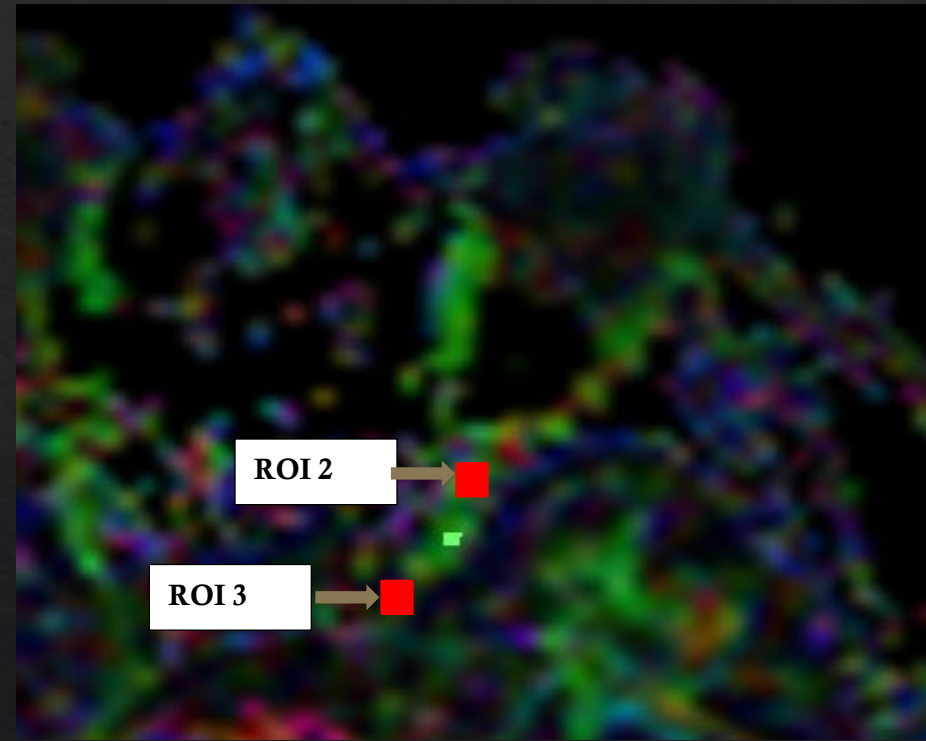
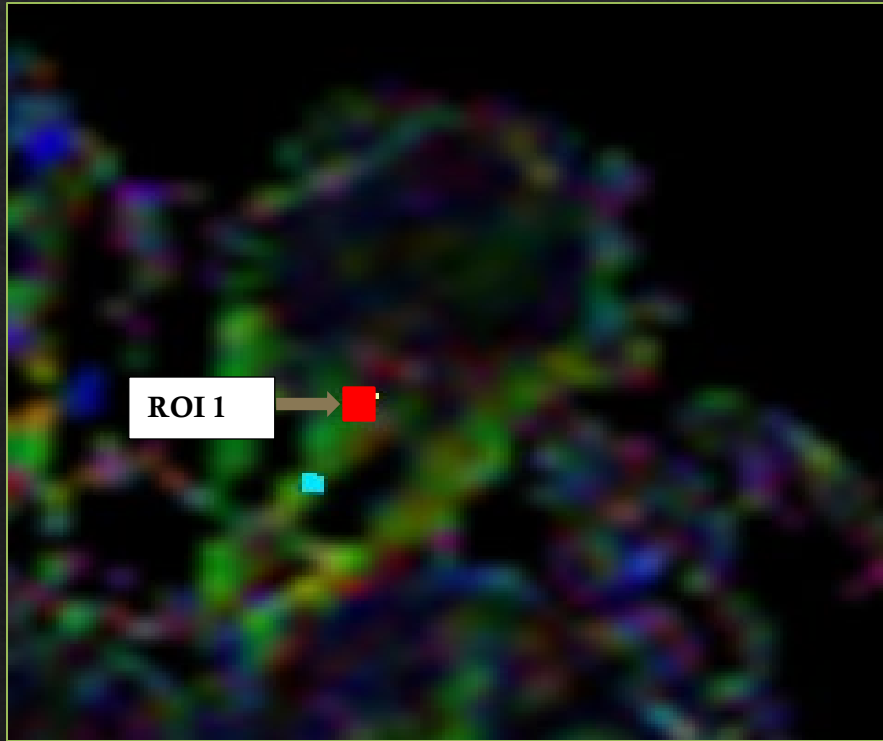


Figure 1b: ROI 4: Lateral geniculate nucleus LGN, ROI 5: directly after/posterior to LGN, ROI 6: at the level of occipital horn of lateral ventricle.

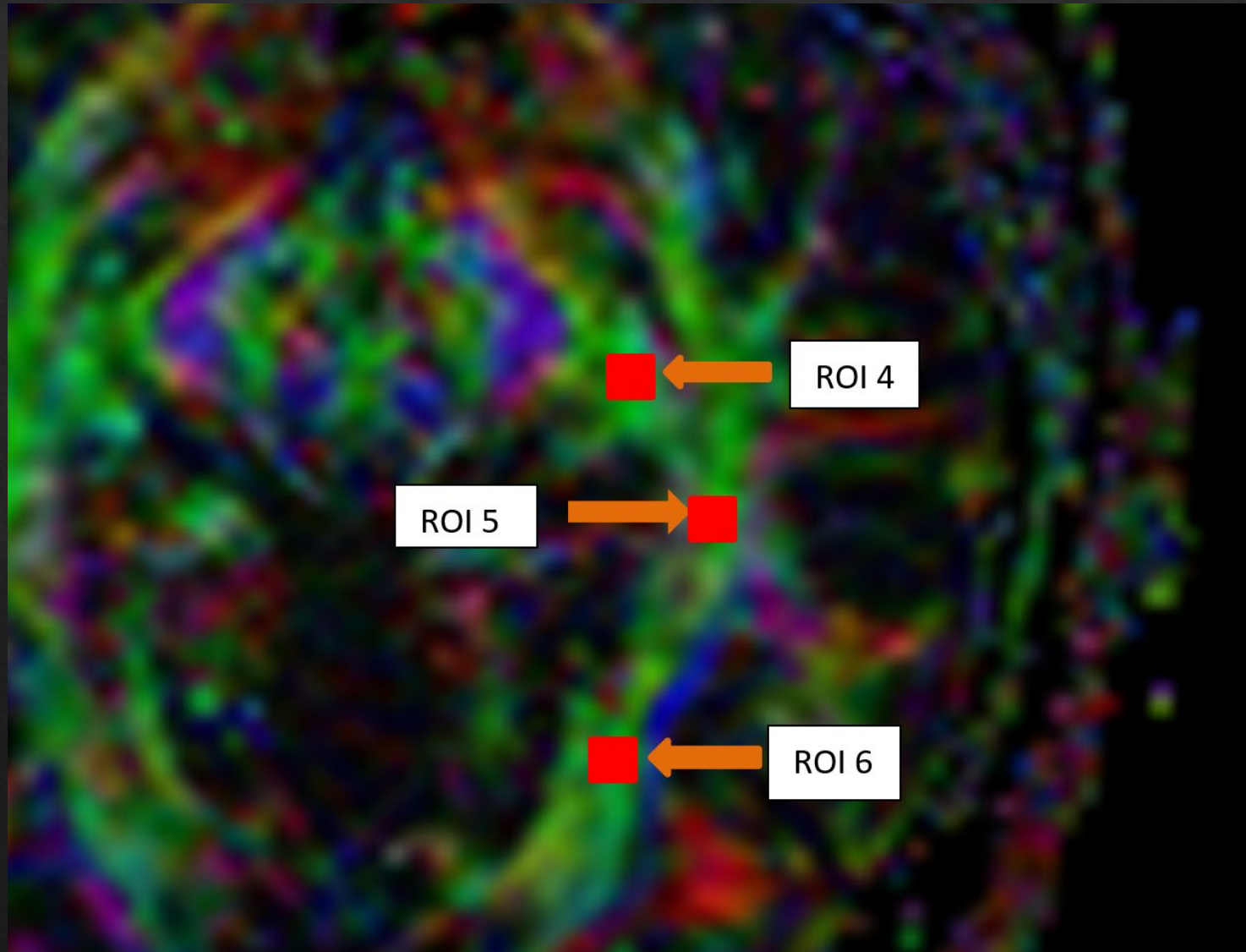
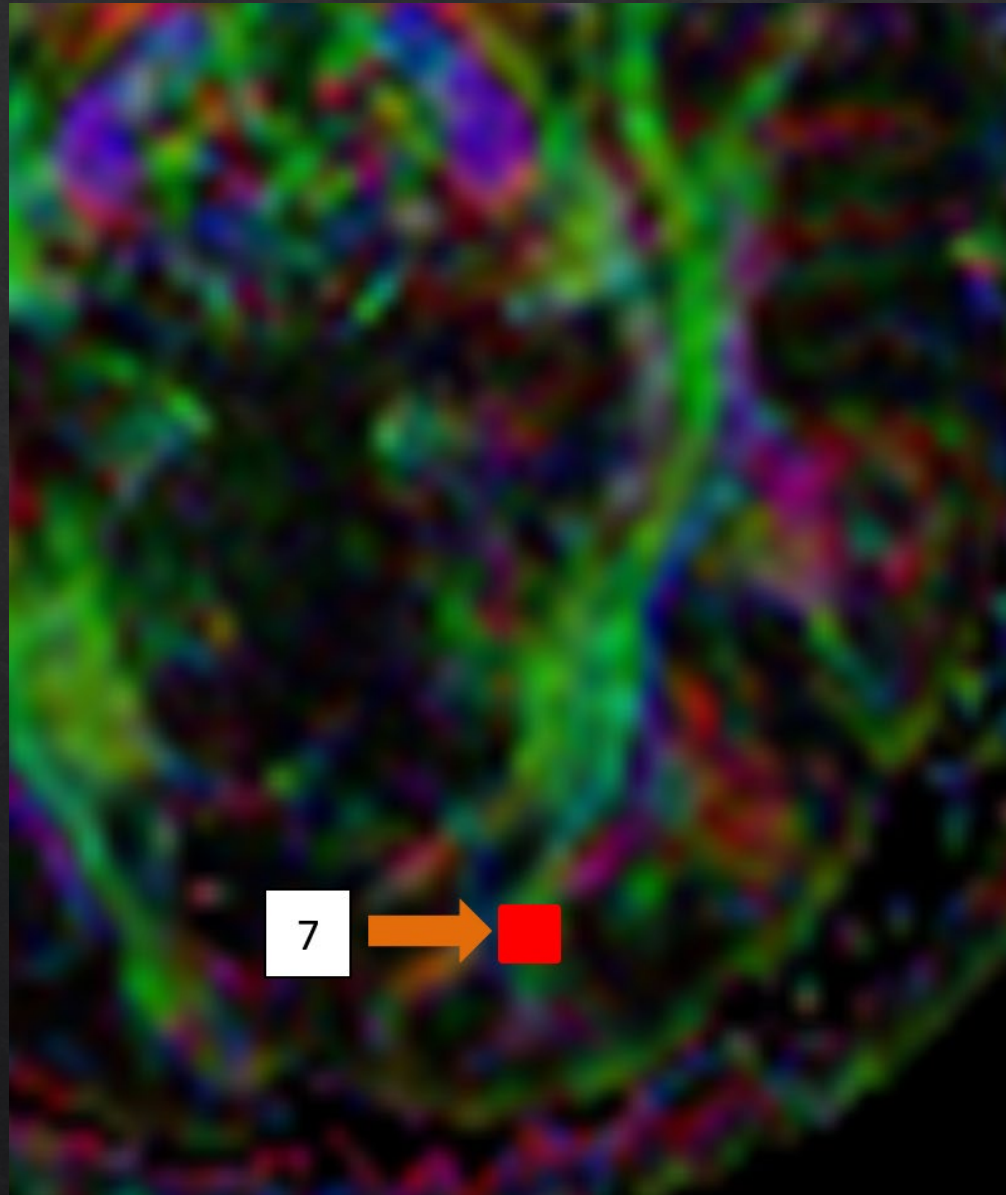


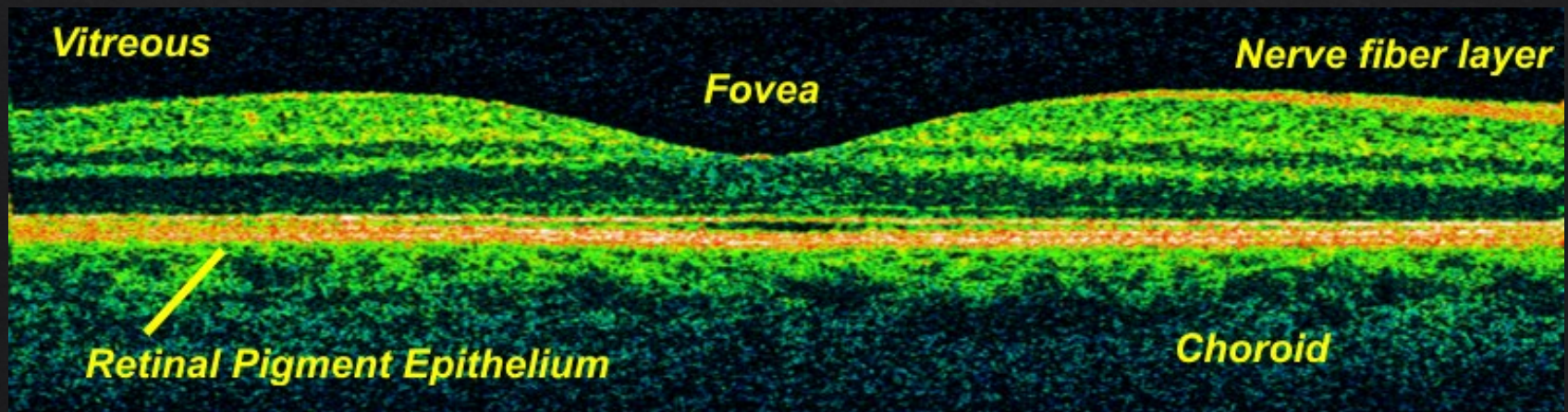
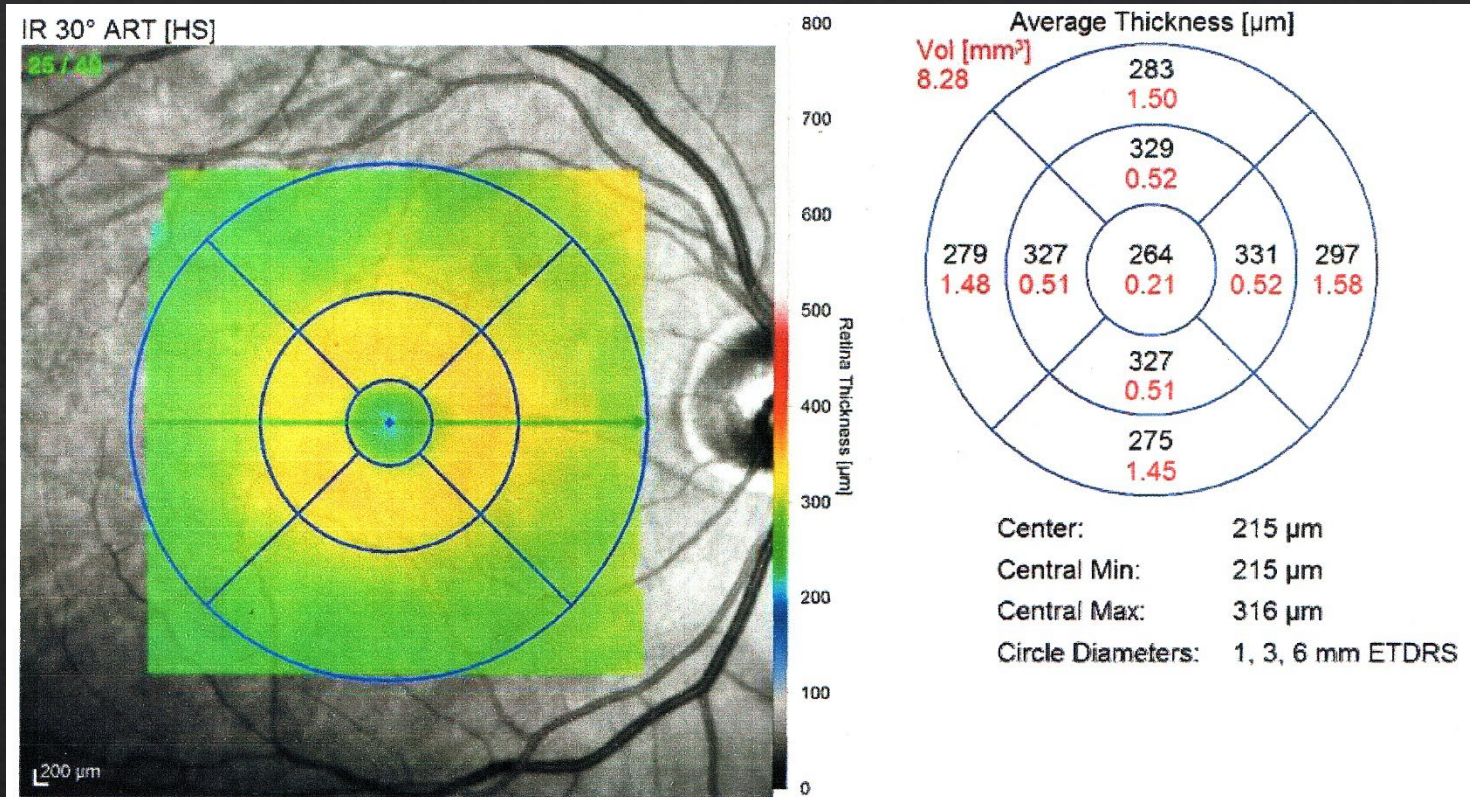
Figure 1c: ROI 7: right before optic fibers cortical spread.



◆ All patients (n = 15) were subjected to extensive ophthalmologic evaluation that included:

- 1) Visual field examination
- 2) Goldmann applanation tonometry to measure IOP
- 3) Optic Coherence Tomography (OCT) to measure the thickness of retinal nerve fiber layer.

OCT map



Exclusion criteria:

- ◆ History of ocular disease/surgery that may affect retinal or optic nerve fibers including optic neuritis, retinal or choroidal detachment, high myopia.
- ◆ MRI-diagnosed white matter disease affecting the visual pathway including demyelinating disorders or old infarcts.

Figure 2: Gender distribution among study subjects

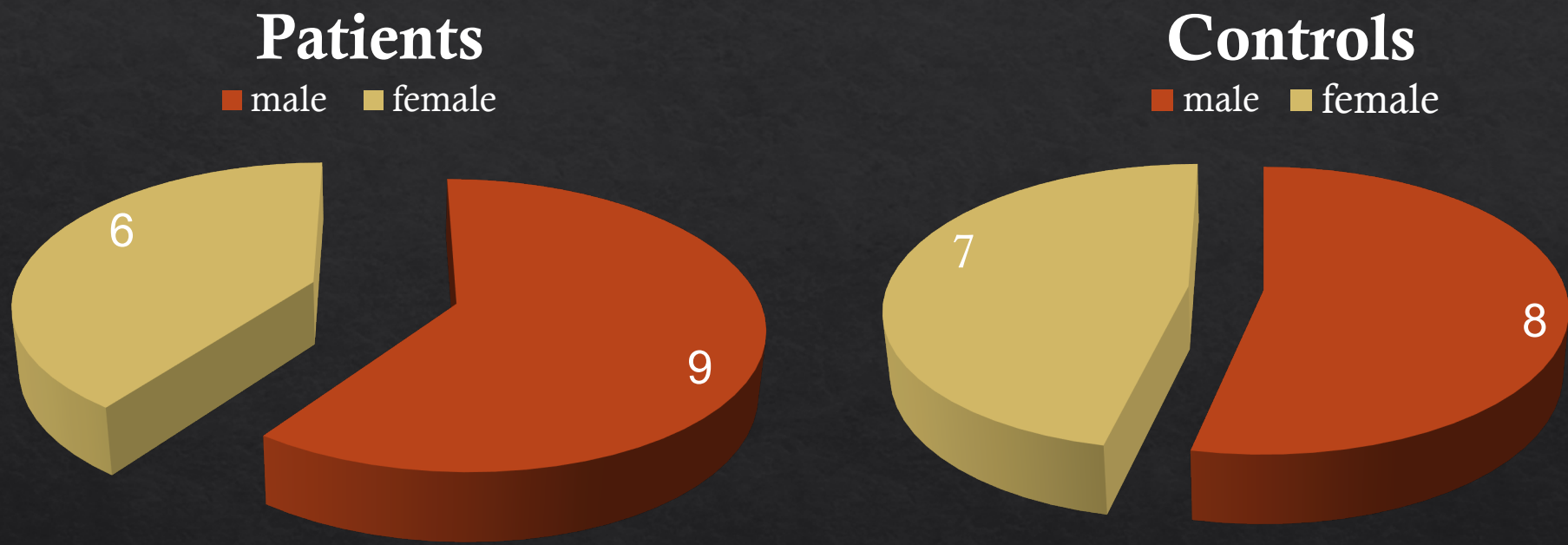


Figure 3: Age characteristics (numbers in years) of study subjects

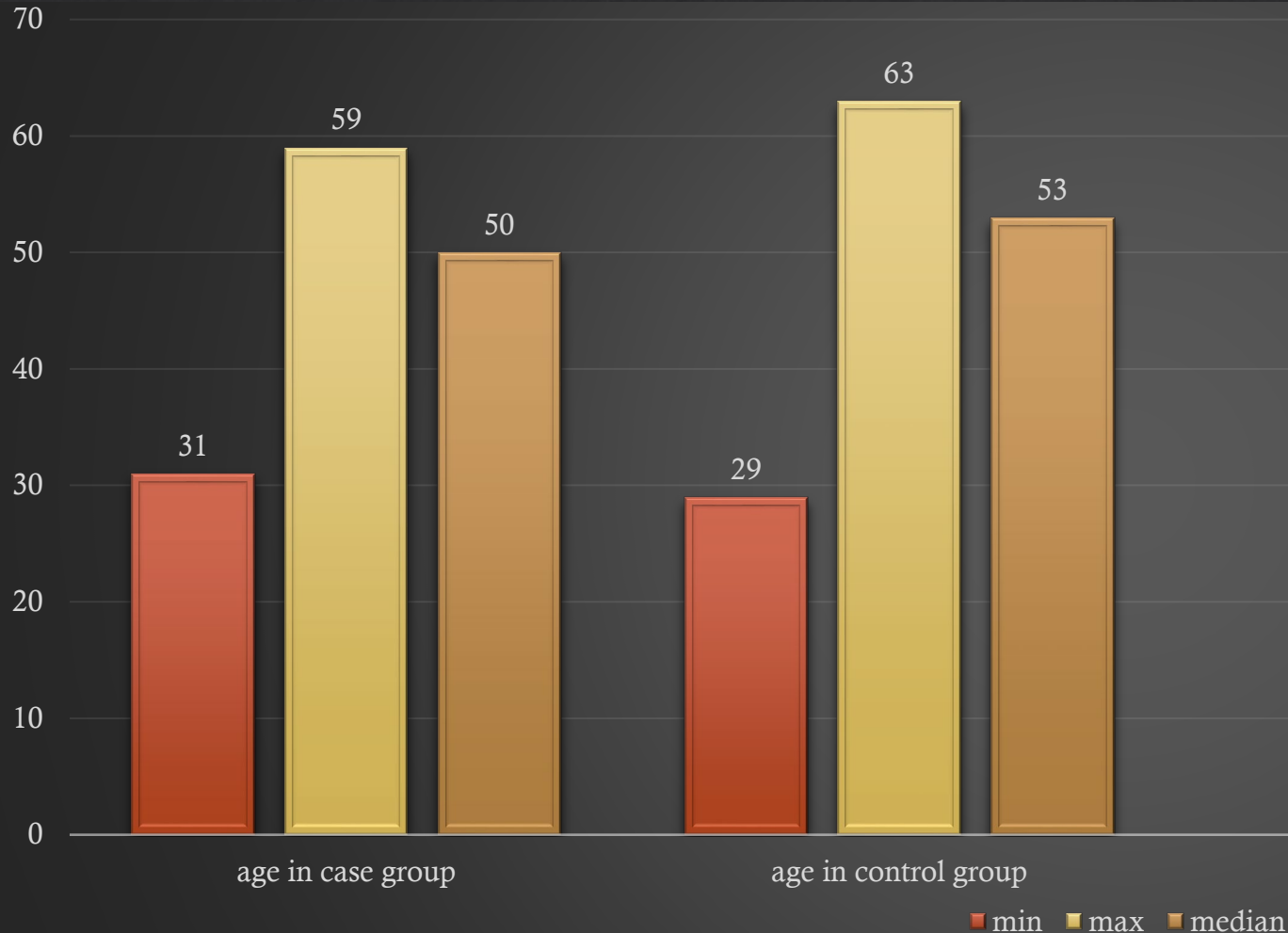
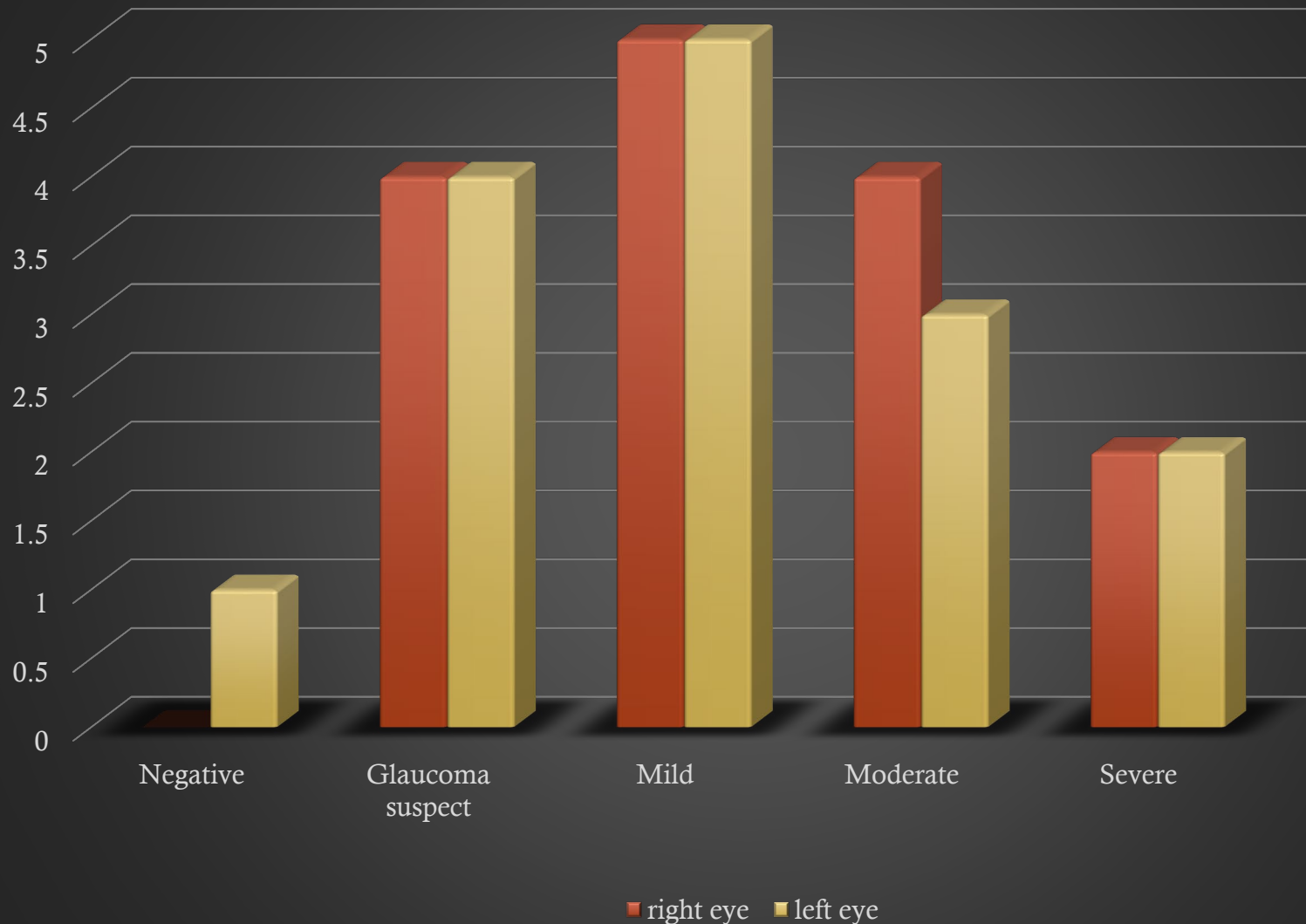


Figure 4: Distribution of cases according to clinical severity of glaucoma in right and left eye



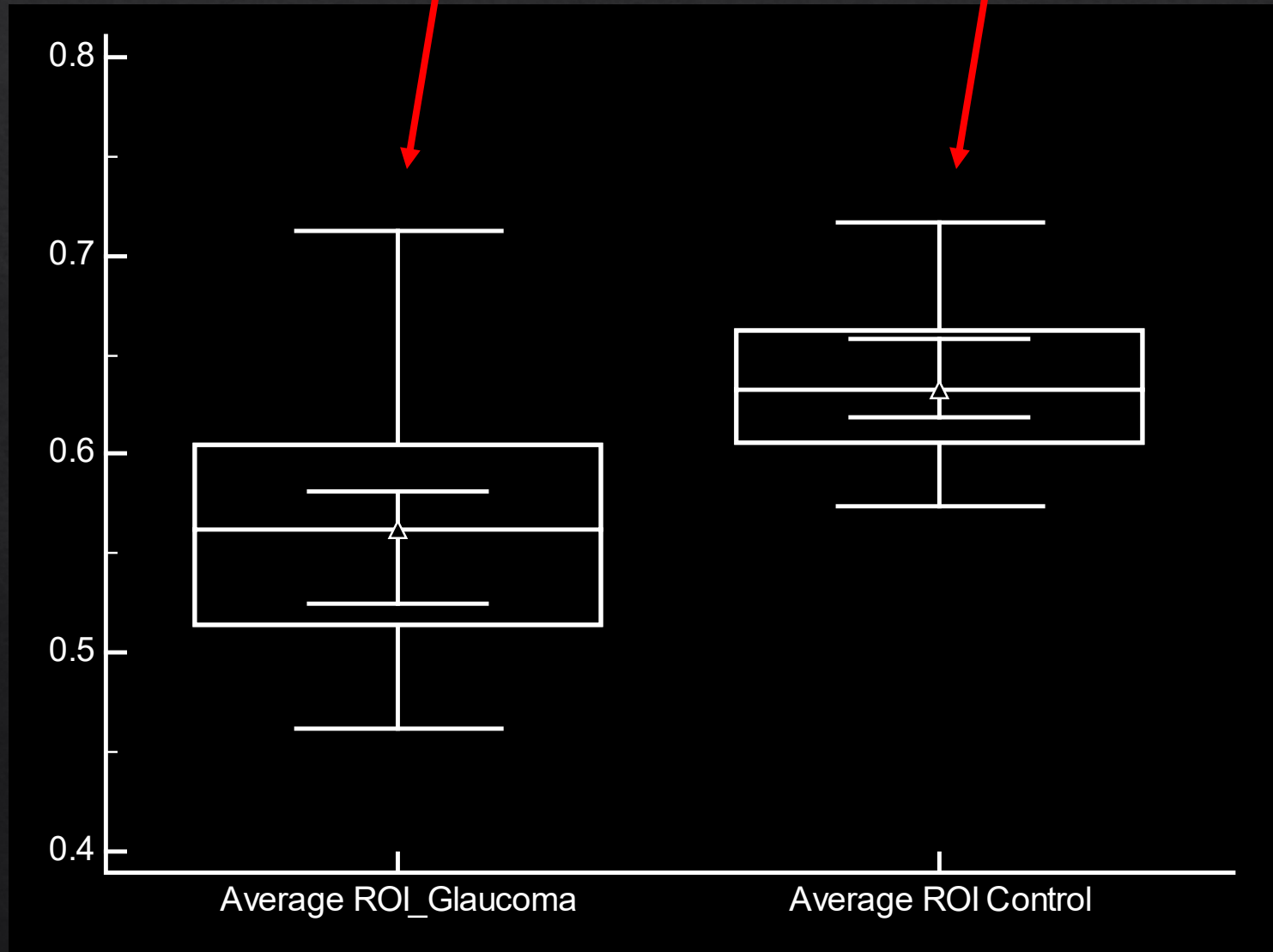
Main DTI findings:

- ◇ FA values were significantly lower at bilateral 7 different segments of the visual pathway in POAG patients compared to normal controls.
 - ◇ *On the Right side: FA = 0.52 (patients) versus 0.63 (controls),
 $p = 0.001$*
 - ◇ *On the Left side: FA = 0.58 (patients) versus 0.63 (controls),
 $p = 0.009$*
- ◇ At the intracranial segment of the optic nerve (ROI 2), there was a significant correlation between FA values and the total RNFL (retinal nerve fiber layer) thickness ($r = 0.7$, $p = 0.004$).
- ◇ At the intracranial segment of the optic nerve (ROI 2), there was a significant negative correlation between FA values and the duration of disease ($r = -0.6$, $p = 0.03$).

Main DTI findings (continued):

- ◇ No statistically-significant results could be obtained in regards to the relationship between FA values and either the intra-ocular pressure or the clinical severity of disease.

Figure 5: Box-Whisker plot comparing the mean FA values of bilateral 7 ROIs: FA (patients) = 0.57, FA (controls) = 0.64, $p < 0.0001$



Conclusion



(1) DTI demonstrated a statistically-significant correlation as lower FA values with glaucomatous changes of the visual pathway.

(2) DTI might be helpful if used as an imaging biomarker of disease progression in patients with primary open angle glaucoma.

Thank you
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